

Sidharth Rony, Ph.D.

United Nations University - MERIT, Boschstraat 24, 6211 AX Maastricht



Education

- 2018 – 2024 **PhD in Economics** University of London, Royal Holloway, United Kingdom.
Research Interests: *Skills, Technology and Labour, Machine Learning.*
- 2019 – 2020 **Doctoral Coursework in Economics of Education** University of Zurich, Switzerland.
- 2014 – 2016 **MSc Economics and Financial Research**, Maastricht University, Netherlands.
Thesis: *Signalling Asymmetric Information in Auctions.*
- 2007 – 2011 **B.Tech. Electronics and Communication Engineering** Amrita Vishwa Vidyapeetham, India.
Thesis: *VLSI Implementation of Neural Network With Genetic Algorithm Learning.*

Employment

Research

- 01.2025 – **Researcher**, United Nations University - MERIT, Maastricht University, Netherlands.
Apply advanced analytics (*network analysis, machine learning, text analysis, LLMs*) to assess how digital and green technologies affect skills demand, employment, and wage inequality in the EU.
- 11.2022 – 12.2024 **Research Assistant**, London School of Economics (LSE), United Kingdom.
Assisted the policy evaluation team at the What Works Centre for Local Economic Growth at the Centre for Economic Performance, LSE.
- 07.2022 – 10.2022 **Research Assistant**, University of London, Royal Holloway, United Kingdom.
Assisted Dr. Ija Trapeznikova in research on public-sector employment and private job creation in Ghana.
- 07.2021 – 10.2021 **Consultant**, Organisation for Economic Co-operation and Development (OECD), France.
Analysed and interpreted job posting data from Lightcast for multiple OECD projects.
- 05.2021 – 07.2021 **Research Assistant**, University of London, Royal Holloway, United Kingdom.
Collected online job advertisement data in the UK and applied NLP for text extraction.
- 01.2018 – 04.2018 **Research Associate**, Centre for Development Studies, Thiruvananthapuram, India.
Led a 14-member team collecting data for the Kerala Migration Survey 2018.
- 01.2016 – 08.2016 **Research Assistant**, Maastricht University, Netherlands.
Assisted Dr. Christian Seel in research on auction theory; responsibilities included literature classification, proofreading, and data cleaning.

Teaching

- 2025 *Natural Language Processing for Social Scientists (PhD, Python)*, Lecturer, UNU-MERIT.
- 2024 – 2025 *Family Economics (UG)*, TA, Royal Holloway, University of London.
- 2019 – 2025 *Econometrics (UG, R)*, TA, Royal Holloway, University of London.
- 2021 – 2022 *Coding and Programming for Economists (UG, Python)*, TA, Royal Holloway, University of London.

Employment (continued)

2018 – 2020 *Data Skills for Economists (UG, R)*, TA, Royal Holloway, University of London.
2018 – 2020 *Applied Economics and Policy (UG)*, TA, Royal Holloway, University of London.

Industry

07.2011 – 06.2014 **Electronics Engineer**, Wipro Ltd., Bangalore/Cochin, India.
C developer in the Product Engineering Service (PES) Department; designed and developed embedded systems for multiple clients.

Research

Peer Reviewed Articles

- **S. Rony**, *LLM Meets Job Advertisements: Unmasking Skill Premia in the UK*, Forthcoming in *Research in Labor Economics*, Vol. 54, Emerald Publishing. (JMP; UNU-MERIT Working Paper 2026-010; also available on SSRN), 2026.  URL: <https://unu-merit.nl/publications/wppdf/2026/wp2026-010.pdf>.

Working Papers

- **S. Rony** and A. Badort, *How Regions Use Large Language Models*, (upon request), 2026.
- **S. Rony**, T. Ciarli, and Ö. Nomaler, *Green and Digital Technology: Exposure, Substitution and Complementarity*, (upon request), 2026.
- **S. Rony**, T. Ciarli, and Ö. Nomaler, *LLM as Survey Agent*, (upon request), 2025.
- **S. Rony** and J. Patman, *Standard Occupation Classifier: A Natural Language Processing Approach*, arXiv:2511.23057, 2025.  URL: <https://arxiv.org/abs/2511.23057>.
- **S. Rony**, *Automation and Human Capital Investment*, SSRN Working Paper No. 5821482, 2024.  URL: <https://ssrn.com/abstract=5821482>.

Work in Progress

- A. Badort, B. Caldarola, T. Ciarli, and **S. Rony**, *Matching Lightcast Skills to the ESCO Framework*, 2026.
- A. Badort, B. Caldarola, T. Ciarli, and **S. Rony**, *Recombining Capabilities: Skill Demand Portfolios and Regional Productivity*, 2026.
- A. Badort, B. Caldarola, T. Ciarli, and **S. Rony**, *Twin Transition Skill Demand and Regional Productivity*, 2026.
- **S. Rony** and A. Badort, *Skill Relatedness and the Absorption of AI Across US Labor Markets*, 2026.
- **S. Rony**, T. Ciarli, A. Marzucchi, U. Rizzo, and A. Vanegas, *The Twin Transition in Labour Demand: Green and Digital Skills in European Online Job Vacancies*, 2026.
- **S. Rony**, T. Ciarli, and Ö. Nomaler, *Green and Digital Technology: Occupational Exposure and Labour Market Effects*, 2026.

Technical Reports

- Data Study Group Team, *Data Study Group Final Report: Defence Science & Security Laboratory (DSTL) Using Topic Modelling to Discover New Trends in the Scientific Literature*, 2023.  DOI: 10.5281/zenodo.10033237.

Conference Proceedings

- V. Vijayakumar, S. Menon, K. Divakaran, and **S. Rony**, *Modified Genetic Algorithm with Controlled Population Dynamics for Efficient Resource Utilization*, Bhubaneswar, India, 2011.

Skills

Languages	English, Malayalam, Hindi, Tamil
Coding	Python, Stata, R, SQL, Matlab, C, C++

Awards and Grants

2022–2024	Adzuna Job Advertisement Data – Awarded data to act as lead researcher analysing COVID-19’s impact on UK skills demand using Adzuna job advertisement data.
2018–2024	Economic and Social Research Council (ESRC) Studentship – PhD scholarship with Professor Arnaud Chevalier on “Ageing, Automatization and the Future of Skills in the Labour Market,” in partnership with the OECD.
2020	Royal Economic Society Small Academic Expenses Grant – Awarded to attend the Doctoral Course at the University of Zurich.
2019	RHUL Travel Grant – Departmental funding awarded to attend the Doctoral Course at the University of Zurich. SeNSS Research and Training Grant – Combination of SeNSS Research and Training Grant and SeNSS Summer School bursaries awarded to attend the Essex Summer School 2019.
2012	PES–Shining Star , Wipro Technologies, India – Awarded for outstanding performance in post-silicon validation.

Certification

2021–2026	Office for National Statistics (ONS) Accredited Researcher.
2023	Associate Fellow of the Higher Education Academy